Separation, Pooling, and Endogenous Timing

A Tullock Contest under Two-sided Asymmetry

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Introduction

Timing is a strategic variable

In labor–management bargaining, lobbying, and litigation, the two sides compete over resources and over when to act.

- Move **early**: commitment — but the action is exposed
- Move **late**: flexibility — but the initiative is lost

When one side knows more

When a party understands its own stake far better than its opponent, the incentives to move early or late are no longer symmetric.

Research question

Does the better-informed side want to commit first, or wait?

Fu (2006) studies a two-player Tullock contest:

- Informed player I privately observes his prize value
- Uninformed player U knows only the prior distribution
- Both first choose *when* to move, then choose effort

Fu's conclusion

Result

The informed player *I* prefers to move **second**.

- Moving first **reveals** *I*'s valuation to *U*
- This erodes the rent from private information
- Waiting lets *I* react after observing *U*, keeping the informational advantage

Where Fu's result comes from

The conclusion rests on a specific **signaling channel**:

1. *I*'s effort is informative about his type
2. *U* updates his posterior belief
3. *U*'s best response shifts with that belief

This channel sustains pooling and makes *waiting* a strict preference.

Our one change

Modification

Let *I*'s type affect **only *I*'s own prize**, and **not** enter *U*'s continuation best response.

- *U*'s posterior about *I* can no longer move *U*'s strategy
- The signaling mechanism breaks down

What this implies

Timing is **no longer a signaling game**.

It reduces to

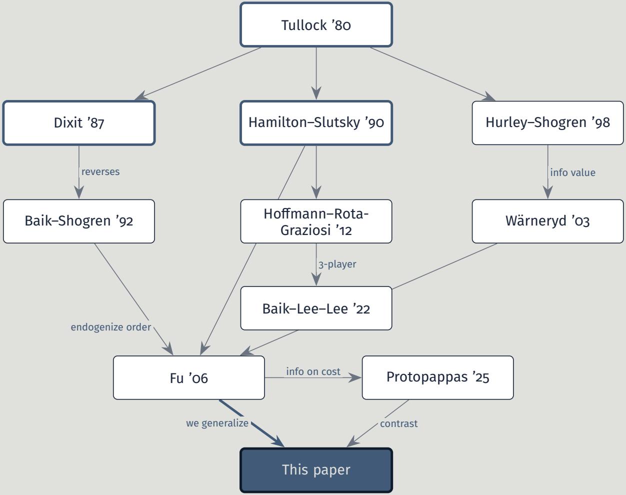
a pure comparison of *ex ante expected payoffs* between moving first and moving second.

Separation will turn out to be a property of the contest *structure* itself.

Main results

1. The IU subgame has a unique separating equilibrium — and this is **distribution-free**.
2. Timing is governed by **ex ante** expected payoffs; type-level gains are only an ex post decomposition.
3. V, μ, δ **jointly** determine the preference — μ versus V alone is not enough.

LITERATURE: HOW THE THREADS CONNECT



Literature: where we stand

Paper	Order	Private info	Position / how we differ
Tullock (1980)	—	—	Contest technology; we adopt the r=1 benchmark
Dixit (1987)	exogenous	none	Commitment value; we endogenize the order
Baik and Shogren (1992)	endogenous	none	Underdog leads; we add a private value
Hamilton and Slutsky (1990)	endogenous	none	The three-subgame template we reuse
Hoffmann and Rota-Graziosi (2012)	endogenous	none	"Weak leads" under substitutes; we revise via correlation
Hurley and Shogren (1998)	exog. (sim.)	prize value	Effort comparison; we compare *timing*
Wärneryd (2003)	exogenous	common value	Info advantage has value; order still fixed
Fu (2006)	endogenous	prize value	Signal → pooling; the perfect-correlation *endpoint*
Protopappas (2025)	endogenous	future cost	Timing signals cost; we close that channel
This paper	endogenous	prize value	Correlation spectrum; new coexistence region (sep. & pool.)

Contribution

- We change *where* the private information sits
- Separation becomes a consequence of the **contest structure**, not of a belief assumption or refinement
- To our knowledge, this distribution-free form of the separation result is new

Model

Players & prize

Two risk-neutral players:

- Agent I — informed
- Agent U — uninformed

They compete over whether a policy is implemented. Winner takes his own prize; loser gets zero.

- I wins $\Rightarrow$ obtains $V_I > 0$
- U wins $\Rightarrow$ obtains $V_U \equiv V > 0$

Two layers of asymmetry

- **Information**
  - U knows the prior G of V_I, never its realization
  - I observes V_I privately — only *after* the timing decision
- **Prize**: I receives V_I, U receives V (no requirement $V_I = V$)

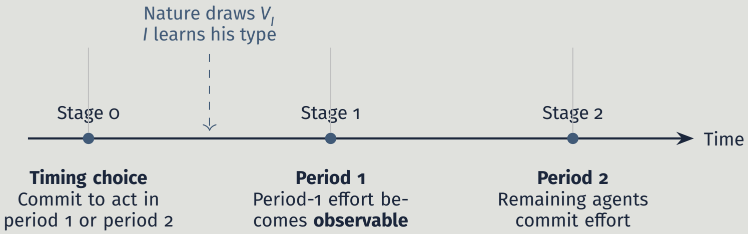
Information structure

1. Stage 0: I and U simultaneously choose *when* to act — I has not yet seen his type
2. Nature draws V_I from G
3. I observes V_I privately; U does not
4. Players enter the effort subgame at their committed times

Consequence

The stage-0 choice is made on **ex ante** expected payoffs.

Timeline of the contest



The three-stage sequential game.

Three regimes

		Agent U Period 1	Agent U Period 2
Agent I	Period 1	SS	IU
	Period 2	UI	SS

- SS — simultaneous moves
- UI — U commits first, I responds
- IU — I commits first, U responds

Which regime can carry a signal?

- SS — simultaneous: no signal
- UI — U moves first: no signal from I
- IU — I moves first, U observes x_I: a signal could exist here

Only IU raises the question of whether I's effort transmits type information.

Following Tullock (1980), efforts $x_I, x_U \geq 0$ give the proportional winning probabilities

$$p_I = \frac{x_I}{x_I + x_U}, \qquad p_U = \frac{x_U}{x_I + x_U}. \tag{1}$$

Payoffs

Net of effort costs, with I of realized type v:

$$\pi_I(x_I, x_U; v) = \frac{x_I}{x_I + x_U} v - x_I,$$

$$\pi_U(x_I, x_U; V) = \frac{x_U}{x_I + x_U} V - x_U.$$

Interior first-order conditions give

$$x_I^*(x_U; v) = \sqrt{v\, x_U} - x_U, \tag{2}$$

$$x_U^*(x_I; V) = \sqrt{V\, x_I} - x_I. \tag{3}$$

Outside the interior, the relevant best response is the maximum of zero and these expressions.

The single fact behind every result

x_U^* depends only on x_I and V — it contains **no posterior belief** about V_I.

A signal may exist at the *observational* level, but it never enters U's behavior.

This is the principal departure from Fu (2006).

Stage-0: an ex ante timing game

Since V_I is not yet realized, I compares $\tilde{\pi}_I^J \equiv \mathbb{E}\left[\pi_I^J(V_I)\right]$, $J \in \{\mathrm{SS}, \mathrm{UI}, \mathrm{IU}\}$.

		Agent U	
		Period 1	Period 2
Agent I	Period 1	$(\tilde{\pi}_I^{\mathrm{SS}}, \pi_U^{\mathrm{SS}})$	$(\tilde{\pi}_I^{\mathrm{IU}}, \pi_U^{\mathrm{IU}})$
	Period 2	$(\tilde{\pi}_I^{\mathrm{UI}}, \pi_U^{\mathrm{UI}})$	$(\tilde{\pi}_I^{\mathrm{SS}}, \pi_U^{\mathrm{SS}})$

Plan: solve the three effort subgames, then read this 2×2 game.

Equilibrium Analysis

Notation: moments of the prior

$$\mu \equiv \mathbb{E}[V_I], \quad \kappa \equiv \mathbb{E}\left[\frac{1}{\sqrt{V_I}}\right], \quad \nu \equiv \mathbb{E}[\sqrt{V_I}], \quad \lambda \equiv \mathbb{E}\left[\frac{1}{V_I}\right]. \tag{4}$$

- All assumed finite; each subgame has an interior solution
- What matters: how timing reorders **commitment**, **observation**, **response**

I tailors effort to type; U chooses one effort under uncertainty.

$$x_U^{SS} = \frac{V^2\kappa^2}{(1+V\lambda)^2}, \tag{5}$$

$$x_I^{SS}(v) = \frac{V\kappa\sqrt{v}}{1+V\lambda} - x_U^{SS}. \tag{6}$$

Simultaneity ≠ symmetric effort.

$$\tilde{\pi}_I^{SS} = \mu - \frac{2V\kappa v}{1 + V\lambda} + \frac{V^2\kappa^2}{(1 + V\lambda)^2},$$

$$\pi_U^{SS} = \frac{V^3\kappa^2\lambda}{(1 + V\lambda)^2}.$$

U's effort absorbs the ex ante uncertainty about *I*'s type.

Proof. See proof in page 73. □

UI: UNINFORMED LEADER — EFFORTS

U commits first; *I* responds type-by-type.

$$x_U^{\mathrm{UI}} = \frac{V^2\kappa^2}{4}, \tag{7}$$

$$x_I^{\mathrm{UI}}(v) = \frac{V\kappa\sqrt{v}}{2} - \frac{V^2\kappa^2}{4}. \tag{8}$$

U's effort is now a **Stackelberg commitment**, not the simultaneous FOC.

UI: EX ANTE PAYOFFS

$$\tilde{\pi}_I^{\mathrm{UI}} = \mu - V\kappa v + \frac{V^2\kappa^2}{4},$$

$$\pi_U^{\mathrm{UI}} = \frac{V^2\kappa^2}{4}.$$

Proof. See proof in page 76. □

I moves first; U responds to x_I — but his best response ignores beliefs.

$$x_I^{\mathrm{IU}}(v) = \frac{v^2}{4V}, \qquad x_U^{\mathrm{IU}}(v) = \frac{v(2V - v)}{4V}. \tag{9}$$

$$\tilde{\pi}_I^{\mathrm{IU}} = \frac{\mathbb{E}[V_I^2]}{4V}.$$

Theorem 1 (Unique separating equilibrium)

If type v mimics any $s \neq v$, the payoff loss is

$$\frac{(v - s)^2}{4V} > 0.$$

Every type strictly prefers its own action ⇒ pooling and semi-separating equilibria are ruled out. Distribution-free.

Proof. See proof in page 78. □

Three regimes side by side

	$\tilde{\pi}_I^J$	π_U^J
SS	$\mu - \frac{2V\kappa v}{1+V\lambda} + \frac{V^2\kappa^2}{(1+V\lambda)^2}$	$\frac{V^3\kappa^2\lambda}{(1+V\lambda)^2}$
UI	$\mu - V\kappa v + \frac{V^2\kappa^2}{4}$	$\frac{V^2\kappa^2}{4}$
IU	$\frac{\mathbb{E}[V_I^2]}{4V}$	$\frac{4V^2-4V\mu+\mathbb{E}[V_I^2]}{4V}$

These feed directly into the stage-0 game.

Endogenous Timing

		Agent U	
		Period 1	Period 2
Agent I	Period 1	$(\tilde{\pi}_I^{SS}, \pi_U^{SS})$	$(\tilde{\pi}_I^{IU}, \pi_U^{IU})$
	Period 2	$(\tilde{\pi}_I^{UI}, \pi_U^{UI})$	$(\tilde{\pi}_I^{SS}, \pi_U^{SS})$

We compare unilateral deviations from this timing game.

Four deviation differences

$$\Delta_I^{UI} \equiv \tilde{\pi}_I^{UI} - \tilde{\pi}_I^{SS}, \qquad \Delta_I^{IU} \equiv \tilde{\pi}_I^{IU} - \tilde{\pi}_I^{SS},$$

$$\Delta_U^{SS} \equiv \pi_U^{SS} - \pi_U^{IU}, \qquad \Delta_U^{UI} \equiv \pi_U^{UI} - \pi_U^{SS}.$$

Each sign tells us whether a player wants to deviate from a candidate profile.

U's deviation when I moves second

$$\Delta_U^{\mathrm{UI}} = \frac{V^2\kappa^2(1 - V\lambda)^2}{4(1 + V\lambda)^2} \;\geq\; 0.$$

Whenever I plans to move second, U generically prefers to move first — equality only at the knife-edge $V\lambda = 1$.

Stage-0 best responses (Lemma)

Lemma 2 (Best responses in the stage-0 timing game)

- *If U chooses period 1, then I chooses period 2 iff* $\Delta_I^{UI} > 0$*; otherwise he chooses period 1.*
- *If U chooses period 2, then I chooses period 1 iff* $\Delta_I^{IU} > 0$*; otherwise he chooses period 2.*
- *If I chooses period 1, then U chooses period 1 iff* $\Delta_U^{SS} > 0$*; otherwise he chooses period 2.*
- *If I chooses period 2, then U weakly prefers period 1 because* $\Delta_U^{UI} \geq 0$.

Proof. See proof in page 82. □

Which regimes are equilibria?

Theorem 3 (Pure-strategy timing equilibria)

- UI $\Leftrightarrow$ $\Delta_I^{UI} \geq 0$
- *Immediate SS* $\Leftrightarrow$ $\Delta_I^{UI} \leq 0$ *and* $\Delta_U^{SS} \geq 0$
- IU $\Leftrightarrow$ $\Delta_I^{IU} \geq 0$ *and* $\Delta_U^{SS} \leq 0$
- *Delayed SS* $\Leftrightarrow$ $\Delta_I^{IU} \leq 0$ *and* $V\lambda = 1$

Proof. See proof in page 83. □

No refinement needed

- Regime selection is read **directly** from the four deviation differences
- Delayed SS needs the knife-edge $V\lambda = 1$
- For generic parameters it cannot be a pure-strategy equilibrium

Why ex ante, not type-level

Define the ex post comparison for a realized type:

$$R(v) \equiv \pi_I^{SS}(v) - \pi_I^{IU}(v) = \left(\sqrt{v} - \frac{V\kappa}{1 + V\lambda}\right)^2 - \frac{v^2}{4V}. \tag{10}$$

Identity

$$\mathbb{E}[R(V_I)] = \tilde{\pi}_I^{SS} - \tilde{\pi}_I^{IU} = -\Delta_I^{IU}.$$

A single type's ex post preference is just a **decomposition**; the stage-0 choice is governed by the *average*.

Proof. See proof in page 80. □

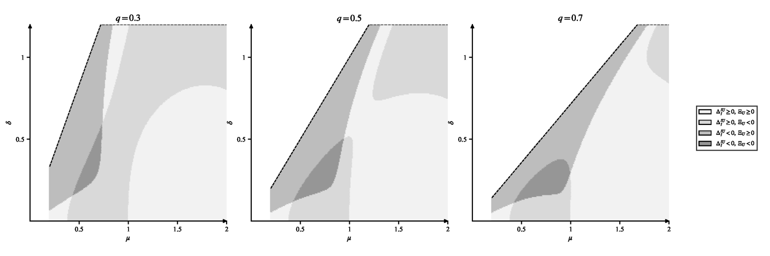
Lemma 4 (Two-point sign test)

Let $V_I \in \{v_L, v_H\}$. Comparing the signs of $R(v_L)$ and $R(v_H)$ gives:

- $(-,-)$: *both types prefer moving first, so* $\Delta_I^{IU} > 0$
- $(+,+)$: *both types prefer moving second, so* $\Delta_I^{IU} < 0$
- *mixed signs: the aggregate sign is determined by the probability-weighted average*

Proof. See proof in page 84. □

Two-point: timing regions



Sign regions on the (μ, δ) plane.

V, μ, δ *jointly* set the timing preference.

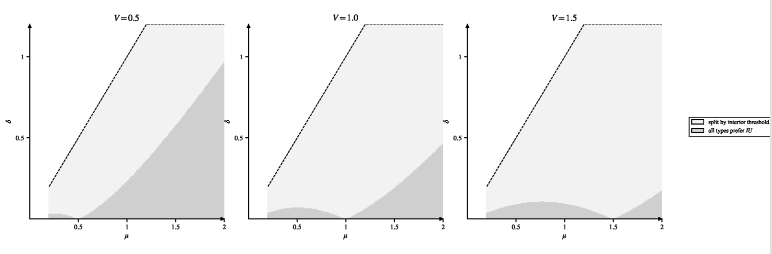
Lemma 5 (Uniform threshold characterization)

If $V_I \sim U[\mu - \delta, \mu + \delta]$, then the continuity of $R(v)$ implies:

- *opposite endpoint signs $\Rightarrow$ an interior threshold v^* where the comparison reverses*
- *same endpoint sign $\Rightarrow$ all types prefer the same direction*

Proof. See proof in page 85. □

Uniform: timing regions



Type-partition regions; panels are V = 0.5, 1.0, and 1.5.

As V rises, the split region expands.

Welfare ≠ equilibrium ranking

Total expected welfare $\mathbb{E}[W] = \tilde{\pi}_I + \pi_U$:

$$\mathbb{E}[W^{\mathrm{UI}}] = \mu - V\kappa v + \frac{1}{2}V^2\kappa^2,$$

$$\mathbb{E}[W^{\mathrm{IU}}] = V - \mu + \frac{\mathbb{E}[V_I^2]}{2V}.$$

Equilibrium checks *unilateral deviations*; welfare *aggregates*. No fixed monotone correspondence.

Proof. See proof in page 86. □

Extension: Partially Correlated Types

The unique separating equilibrium relied on U's prize being a known constant V.

Now

Let U's prize also be random, $V_U \in \{V_L, V_H\}$, and **positively correlated** with V_I via a single parameter $\rho \in [0, 1]$.

Partial correlation structure

Single parameter $\rho \in [0, 1]$: Pearson correlation between V_I and V_U.
Diagonal mass of the joint distribution:

$$\Pr(V_I = V_H, V_U = V_H) = q^2 + \rho\, q(1 - q). \tag{11}$$

Conditional probabilities are clean linear functions of ρ:

$$\begin{aligned} p_H &\equiv \Pr(V_U = V_H \mid V_I = V_H) = q + \rho(1 - q), \\ p_L &\equiv \Pr(V_U = V_L \mid V_I = V_L) = 1 - q(1 - \rho). \end{aligned}$$

Three benchmarks

- $\rho = 0$: statistical independence — the baseline model ($V_U \equiv V$)
- $\rho = 1$: full correlation — exactly Fu (2006)
- $0 < \rho < 1$: partial correlation — the focus of this chapter

U's effective prize is no longer constant. With posterior $m = \Pr(V_I = V_H \mid x_I)$:

$$\hat{V}_U(m) = V_L + \big[q + \rho(m - q)\big]\Delta V, \qquad \Delta V = V_H - V_L. \tag{12}$$

Key: $\hat{V}_U(q) = qV_H + (1 - q)V_L$ is *independent of ρ* — the SS payoffs are unaffected by correlation.

Sensitivity to beliefs

$$\frac{d\hat{V}_U}{dm} = \rho\,\Delta V. \tag{13}$$

- $\rho = 0$: derivative = 0 — beliefs carry no value
- $\rho > 0$: a higher posterior raises U's expected own prize proportionally to ρ

Mechanism

A high-effort signal means a **stronger opponent** *and* a **higher own prize**.

The “higher stakes” effect offsets deterrence: under positive correlation, x_U^* is **increasing** in m. U no longer retreats from a high signal.

A numerical illustration

Fix $V_H = 2$, $V_L = 1$, $q = 0.5$, observed $x_I = 1$.

- Independence ($\rho = 0$): $\hat{V}_U \equiv 1.5$, $x_U^* = \sqrt{1.5} - 1 \approx 0.225$
- Positive corr. ($\rho = 0.6$): $\hat{V}_U(0) = 1.2$, $\hat{V}_U(1) = 1.8$; x_U^* runs $0.095 \to 0.342$ — more than tripling

When does separation survive?

Only the high type's incentive constraint binds:

$$\frac{\hat{V}_U(1)}{\hat{V}_U(0)} \leq R \equiv \frac{V_H^2}{V_L(2V_H - V_L)}. \tag{14}$$

The low type never gains from upward mimicry — its constraint holds automatically.

THE CRITICAL CORRELATION ρ^*

Because $\hat{V}_U(1)/\hat{V}_U(0)$ is strictly increasing in ρ, the separation condition translates to a unique critical value:

$$\rho^* = \frac{(R-1)\,\hat{V}_U(q)}{\Delta V\,(1-q+Rq)}, \qquad R = \frac{V_H^2}{V_L(2V_H - V_L)}. \tag{15}$$

Undistorted separation holds iff $\rho \leq \rho^*$.

Proof. See proof in page 87. □

Riley distorted separation ($\rho > \rho^*$)

When $\rho > \rho^*$, undistorted separation fails. But low type can **distort effort downward** to block mimicry.

D1 refinement selects the unique distorted equilibrium

$$x_L^{\text{Riley}} = (z_H^-)^2 = \frac{V_H^2}{4\hat{V}_U(0)}\left(1 - \sqrt{\alpha_{\text{sep}}}\right)^2, \quad \alpha_{\text{sep}} \equiv 1 - \frac{\hat{V}_U(0)}{\hat{V}_U(1)}.$$

This is the minimum-cost separating equilibrium (Riley (1979)).

At baseline $V_H = 2V_L$: Riley distorted separation coexists with pooling for *all* $\rho \in (\rho^*, 1]$.

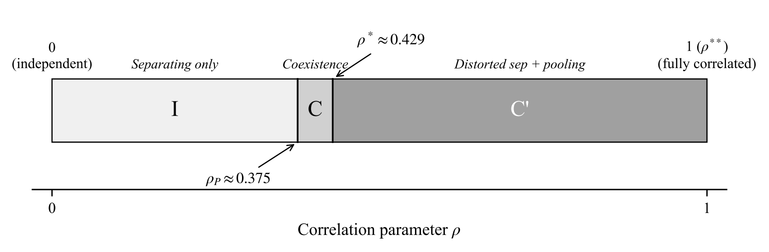
Pooling steps in

When separation collapses, a pooling equilibrium also supports play. With $\alpha \equiv 1 - \hat{V}_U(q)/\hat{V}_U(1)$, it exists iff

$$\sqrt{\alpha} \;\geq\; \frac{V_H - V_L}{V_H + V_L}. \tag{16}$$

Pooling appears at $\rho = \rho_P < \rho^*$. At $\rho = 0$: $\alpha = 0$ — no pooling, consistent with unique separation at independence.

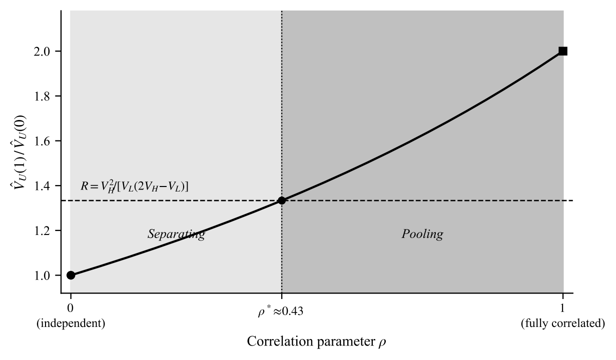
Four equilibrium regions



Equilibrium regions along the $\rho \in [0,1]$ spectrum.

I: sep. only **C**: undist. sep.+pool. **C'**: Riley+pool. **II**: pool. only (empty at $V_H = 2V_L$)

One continuous spectrum



Effective prize ratio $\hat{V}_U(1)/\hat{V}_U(0)$ crosses R uniquely at ρ^*.

Along the ρ-axis the ratio rises monotonically from 1 (at $\rho = 0$) to V_H/V_L (at $\rho = 1$).

Proposition

For $\rho \in [0, \rho^*]$, I's gain from moving first, $\Delta_I^{IU}(\rho)$, is **strictly decreasing** in ρ.

- $\rho = 0$: baseline — I may prefer to move first
- $\rho = 1$: exactly Fu (2006) — I strictly prefers to wait
- Unique crossing $\rho^\dagger \in (0, 1]$

Fu's "informed player always waits" is the $\rho = 1$ endpoint of this spectrum.

Unifying message

Spectrum

Baseline and Fu (2006) are the **two endpoints** of one correlation spectrum.

Fu's pooling outcome is the *degenerate extreme* of the present framework, not a separate world.

Conclusion

What we showed

- Relocating the private information closes the signal at the effort level
- The IU separating equilibrium is then distribution-free
- Timing is an **ex ante** payoff comparison; V, μ, δ act jointly
- Correlation rebuilds a smooth separating $\rightarrow$ pooling spectrum

Takeaway

Whether separation holds is about *where* private information sits — not about belief refinements.

Thank you.

Questions & discussion

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Appendix: Proofs

SS subgame I

Proof. For type v, I's best response to x_U is

$$x_I(v) = \sqrt{v\, x_U} - x_U.$$

Anticipating this type-contingent response, U chooses x_U from the first-order condition

$$V\,\mathbb{E}\left[\frac{x_I(V_I)}{\left(x_I(V_I) + x_U\right)^2}\right] = 1.$$

Since $x_I(v) + x_U = \sqrt{v\, x_U}$, the bracket simplifies and

$$V\,\mathbb{E}\left[\frac{\sqrt{V_I x_U} - x_U}{V_I x_U}\right] = V\left(\frac{\kappa}{\sqrt{x_U}} - \lambda\right) = 1,$$

hence

$$\sqrt{x_U^{SS}} = \frac{V\kappa}{1 + V\lambda}, \qquad x_U^{SS} = \frac{V^2\kappa^2}{(1 + V\lambda)^2}.$$

SS subgame II

Substituting back,

$$x_I^{SS}(v) = \frac{V\kappa\sqrt{v}}{1 + V\lambda} - \frac{V^2\kappa^2}{(1 + V\lambda)^2}.$$

Because $x_I^{SS}(v) + x_U^{SS} = \frac{V\kappa\sqrt{v}}{1+V\lambda}$,

$$\pi_I^{SS}(v) = \left(1 - \frac{V\kappa}{(1 + V\lambda)\sqrt{v}}\right)v - x_I^{SS}(v) = \left(\sqrt{v} - \frac{V\kappa}{1 + V\lambda}\right)^2,$$

$$\pi_U^{SS}(v) = \frac{V^2\kappa}{(1 + V\lambda)\sqrt{v}} - \frac{V^2\kappa^2}{(1 + V\lambda)^2}.$$

SS SUBGAME III

Taking expectations over V_I gives the ex ante payoffs

$$\tilde{\pi}_I^{SS} = \mu - \frac{2V\kappa v}{1 + V\lambda} + \frac{V^2\kappa^2}{(1 + V\lambda)^2},$$

$$\pi_U^{SS} = \frac{V^2\kappa}{1 + V\lambda}\kappa - \frac{V^2\kappa^2}{(1 + V\lambda)^2} = \frac{V^3\kappa^2\lambda}{(1 + V\lambda)^2}.$$

□

UI SUBGAME I

Proof. Given x_U and type v, I's best response is $x_I(v) = \sqrt{v\, x_U} - x_U$. Committing first, U solves

$$\max_{x_U \geq 0} \mathbb{E}\left[V\sqrt{\frac{x_U}{V_I}} - x_U\right] = \max_{x_U \geq 0} \left(V\sqrt{x_U}\,\kappa - x_U\right).$$

The first-order condition $\frac{V\kappa}{2\sqrt{x_U}} - 1 = 0$ yields

$$x_U^{\mathrm{UI}} = \frac{V^2\kappa^2}{4}, \qquad x_I^{\mathrm{UI}}(v) = \frac{V\kappa\sqrt{v}}{2} - \frac{V^2\kappa^2}{4}.$$

Since $x_I^{\mathrm{UI}}(v) + x_U^{\mathrm{UI}} = \frac{V\kappa\sqrt{v}}{2}$,

$$\pi_I^{\mathrm{UI}}(v) = \left(\sqrt{v} - \frac{V\kappa}{2}\right)^2, \qquad \pi_U^{\mathrm{UI}}(v) = \frac{V^2\kappa}{2\sqrt{v}} - \frac{V^2\kappa^2}{4}.$$

UI subgame II

Taking expectations,

$$\tilde{\pi}_I^{\mathrm{UI}} = \mu - V\kappa v + \frac{V^2\kappa^2}{4}, \qquad \pi_U^{\mathrm{UI}} = \frac{V^2\kappa}{2}\kappa - \frac{V^2\kappa^2}{4} = \frac{V^2\kappa^2}{4}.$$

$\square$

Proof. Given x_I, U's best response is $x_U^*(x_I; V) = \sqrt{V x_I} - x_I$. Anticipating it, type v maximizes

$$W(x_I; v) = \frac{x_I}{x_I + x_U^*(x_I; V)} v - x_I = \frac{v}{\sqrt{V}} \sqrt{x_I} - x_I,$$

which is strictly concave. The first-order condition $\frac{v}{2\sqrt{V x_I}} - 1 = 0$ gives

$$x_I^{\mathrm{IU}}(v) = \frac{v^2}{4V}, \qquad x_U^{\mathrm{IU}}(v) = \sqrt{V \frac{v^2}{4V}} - \frac{v^2}{4V} = \frac{v\,(2V - v)}{4V}.$$

Hence

$$\pi_I^{\mathrm{IU}}(v) = \frac{v^2}{4V}, \qquad \pi_U^{\mathrm{IU}}(v) = \frac{(2V - v)^2}{4V},$$

and, in expectation,

$$\tilde{\pi}_I^{\mathrm{IU}} = \frac{\mathbb{E}\left[V_I^2\right]}{4V}, \qquad \pi_U^{\mathrm{IU}} = \frac{4V^2 - 4V\mu + \mathbb{E}\left[V_I^2\right]}{4V}.$$

IU SUBGAME II

Incentive compatibility. If type v mimics any $s \neq v$ it exerts $s^2/(4V)$; after U's response the deviation payoff is

$$\tilde{U}(v,s) = \frac{v}{\sqrt{V}}\sqrt{\frac{s^2}{4V}} - \frac{s^2}{4V} = \frac{2vs - s^2}{4V}.$$

The truthful payoff is $v^2/(4V)$, so

$$\frac{v^2}{4V} - \tilde{U}(v,s) = \frac{(v-s)^2}{4V} > 0 \qquad (s \neq v).$$

Every type strictly prefers its own action; pooling and semi-separating profiles violate incentive compatibility, so the perfect Bayesian equilibrium of the IU subgame is the unique separating one. $\square$

Ex post decomposition $R(v)$ I

Proof. By definition,

$$R(v) = \pi_I^{SS}(v) - \pi_I^{IU}(v) = \left(\sqrt{v} - \frac{V\kappa}{1 + V\lambda}\right)^2 - \frac{v^2}{4V}.$$

Taking expectations over V_I,

$$\mathbb{E}\left[R(V_I)\right] = \tilde{\pi}_I^{SS} - \tilde{\pi}_I^{IU} = -\Delta_I^{IU}.$$

Next, $R(v) \leq 0$ is equivalent to

$$\left(\sqrt{v} - \frac{V\kappa}{1 + V\lambda}\right)^2 \leq \frac{v^2}{4V} \iff \left|\sqrt{v} - \frac{V\kappa}{1 + V\lambda}\right| \leq \frac{v}{2\sqrt{V}}.$$

Since $v/(2\sqrt{V}) > 0$, this absolute-value inequality rewrites as

$$\frac{V\kappa}{1 + V\lambda} \in \left[\sqrt{v} - \frac{v}{2\sqrt{V}},\ \sqrt{v} + \frac{v}{2\sqrt{V}}\right].$$

Ex post decomposition $R(v)$ II

For two-point support, compare the signs of $R(v_L)$ and $R(v_H)$. For the uniform case, continuity and the intermediate value theorem yield an interior root that partitions the support. $\square$

Stage-0 best responses

Proof. Fix the opponent's timing choice and compare the two cells in the stage-0 payoff table. If U chooses period 1, then I compares $\tilde{\pi}_I^{UI}$ against $\tilde{\pi}_I^{SS}$, so period 2 is optimal iff $\Delta_I^{UI} = \tilde{\pi}_I^{UI} - \tilde{\pi}_I^{SS} > 0$. If U chooses period 2, then I compares $\tilde{\pi}_I^{IU}$ against $\tilde{\pi}_I^{SS}$, so period 1 is optimal iff $\Delta_I^{IU} > 0$.

Symmetrically, if I chooses period 1, then U compares π_U^{SS} against π_U^{IU}, so period 1 is optimal iff $\Delta_U^{SS} = \pi_U^{SS} - \pi_U^{IU} > 0$. If I chooses period 2, then U compares π_U^{UI} against π_U^{SS}, and period 1 is weakly optimal exactly when $\Delta_U^{UI} \geq 0$. $\square$

Pure-strategy timing equilibria I

Proof. A pure-strategy equilibrium is a cell in the stage-0 timing game in which both players best respond.

- UI: by the best-response lemma, this cell is an equilibrium exactly when I weakly prefers period 2 after U chooses period 1, i.e. when $\Delta_I^{UI} \geq 0$.
- Immediate SS: this requires that I does not deviate to period 2 when U chooses period 1 and that U does not deviate to period 2 when I chooses period 1, so $\Delta_I^{UI} \leq 0$ and $\Delta_U^{SS} \geq 0$.
- IU: this requires that I weakly prefers period 1 when U chooses period 2 and that U weakly prefers period 2 when I chooses period 1, hence $\Delta_I^{IU} \geq 0$ and $\Delta_U^{SS} \leq 0$.
- Delayed SS: this requires $\Delta_I^{IU} \leq 0$ and $\Delta_U^{UI} \leq 0$. Since $\Delta_U^{UI} \geq 0$ always, the latter holds iff $\Delta_U^{UI} = 0$, i.e. iff $V\lambda = 1$.

□

Two-point sign test

Proof. When V_I has support $\{v_L, v_H\}$ with probabilities $1 - q$ and q,

$$\mathbb{E}[R(V_I)] = (1 - q)R(v_L) + qR(v_H).$$

Since $\mathbb{E}[R(V_I)] = -\Delta_I^{IU}$ by the decomposition proof, if both endpoint values are negative then $\mathbb{E}[R(V_I)] < 0$ and thus $\Delta_I^{IU} > 0$. If both are positive then $\mathbb{E}[R(V_I)] > 0$ and $\Delta_I^{IU} < 0$.
If the endpoint signs differ, then the sign of $\mathbb{E}[R(V_I)]$ is not pinned down pointwise; it is determined by the probability-weighted average of the two endpoint values. $\square$

Uniform threshold characterization

Proof. Under $V_I \sim U[\mu - \delta, \mu + \delta]$, the function $R(v)$ is continuous on the compact support. If the endpoint values have opposite signs, the intermediate value theorem implies the existence of some $v^* \in (\mu - \delta, \mu + \delta)$ such that $R(v^*) = 0$, so preferences reverse at an interior threshold.

If both endpoints have the same sign, continuity rules out any sign change on the interval, and all types therefore prefer the same timing direction. $\square$

Welfare expressions

Proof. Total expected welfare in regime J is $\mathbb{E}[W^J] = \tilde{\pi}_I^J + \pi_U^J$. Summing the ex ante payoffs,

$$\begin{aligned}
\mathbb{E}\left[W^{\mathrm{SS}}\right] &= \tilde{\pi}_I^{\mathrm{SS}} + \pi_U^{\mathrm{SS}} \\
&= \mu - \frac{2V\kappa v}{1 + V\lambda} + \frac{V^2\kappa^2}{(1 + V\lambda)^2} + \frac{V^3\kappa^2\lambda}{(1 + V\lambda)^2},
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}\left[W^{\mathrm{UI}}\right] &= \tilde{\pi}_I^{\mathrm{UI}} + \pi_U^{\mathrm{UI}} = \left(\mu - V\kappa v + \frac{V^2\kappa^2}{4}\right) + \frac{V^2\kappa^2}{4} \\
&= \mu - V\kappa v + \frac{1}{2}V^2\kappa^2,
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}\left[W^{\mathrm{IU}}\right] &= \tilde{\pi}_I^{\mathrm{IU}} + \pi_U^{\mathrm{IU}} = \frac{\mathbb{E}\left[V_I^2\right]}{4V} + \frac{4V^2 - 4V\mu + \mathbb{E}\left[V_I^2\right]}{4V} \\
&= V - \mu + \frac{\mathbb{E}\left[V_I^2\right]}{2V}.
\end{aligned}$$

$\square$

Critical correlation ρ^* I

Proof. Set the high-type IC with equality: $\hat{V}_U(1)/\hat{V}_U(0) = R$, where $R = V_H^2/[V_L(2V_H - V_L)]$. Substituting

$$\hat{V}_U(1) = V_L + [q + \rho(1-q)]\Delta V, \qquad \hat{V}_U(0) = V_L + q(1-\rho)\Delta V,$$

the equality $\hat{V}_U(1) = R\,\hat{V}_U(0)$ expands to

$$V_L + [q + \rho(1-q)]\Delta V = R\big(V_L + q(1-\rho)\Delta V\big).$$

Collecting ρ terms:

$$\rho\big[(1-q) + Rq\big]\Delta V = (R-1)(V_L + q\Delta V) = (R-1)\,\hat{V}_U(q).$$

Dividing both sides by $[(1-q) + Rq]\Delta V > 0$ gives

$$\rho^* = \frac{(R-1)\,\hat{V}_U(q)}{\Delta V\,(1-q+Rq)}.$$

Since $\hat{V}_U(1)/\hat{V}_U(0)$ is strictly increasing in ρ (numerator increases, denominator decreases), undistorted separation holds iff $\rho \le \rho^*$. Two endpoints confirm the regions:

Critical correlation ρ^* II

- $\rho = 0$: $\hat{V}_U(1) = \hat{V}_U(0) = \hat{V}_U(q)$, ratio = $1 < R$ (strict separation).
- $\rho = 1$: $\hat{V}_U(1) = V_H$, $\hat{V}_U(0) = V_L$, ratio = $V_H/V_L > R$ (separation fails — Fu (2006)).

$\square$